CCC-C-426
APRIL 16, 1957
SUPERSEDING
Fed. Spec. CCC-D-651
June 28, 1932

FEDERAL SPECIFICATION

CLOTH, COTTON, DRILL

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope.—This specification covers the requirements for cotton drill cloth used in the manufacture of clothing and equipage items.

1.2 Classification.

1.2.1 Types and classes.—The cloth covered by this specification shall be of the following types and classes as specified (see 6.1):

Type I.—30 inches* 72 by 60, 2.50 yd. (Commercial designation).

Type II.—30 inches* 72 by 48, 2.50 yd. (Commercial designation).

Type III.—30 inches* 72 by 48, 2.85 yd. (Commercial designation).
Class 1.—Greige.
Class 2.—Desized and preshrunk.
Class 3.—Bleached white or dyed and preshrunk.

* Prorata widths of like construction will be acceptable.

2. APPLICABLE SPECIFICATIONS AND STANDARDS

2.1 The following specifications and standards, of the issues in effect on date of invitations for bids, form a part of this specification:

Federal Specifications:

V-T-276—Thread; Cotton.

CCC-T-191—Textile Test Methods.

PPP-P-51 — Packaging, Packing, and Marking of Textile Fabrics (Woolens, Worsteds, Cottons, Silks and Synthetic Fabrics).

Federal Standards:

No. 4 — Glossary of Fabric Imperfections.

No. 102—Preservation, Packaging, and Packing Levels.

(Activities outside the Federal Government may obtain copies of Federal Specifications and Standards as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D. C.

(Federal Government activities may obtain copies of Federal Specifications and Standards and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standards:

MIL-STD-105 — Sampling Procedures and Tables for Inspection by Attributes and Appendix.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

FSC Class 8305

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:29Z
Check the source to verify that this is the current version before use.

3. REQUIREMENTS

3.1 Standard sample. — Unless otherwise specified (see 6.1), the cloth shall be equal to the standard sample in all respects not specifically covered by this specification.

3.2 Preproduction sample approval.—When a preproduction sample approval is required, it shall be so specified in the invitation for bids or the procurement document (see 6.1).

3.3 Material.

3.3.1 *Cotton.*—The cotton shall be of suitable length of staple and grade to meet the requirements of this specification.

3.3.2 *Yarn.*—The cotton used in the manufacture of the yarn shall be thoroughly cleaned, well carded, drawn, and evenly spun into single yarn for both the warp and filling.

3.4 Color.—The color for class 3 cloth shall be as specified (see 6.1 and 6.2) and when colored other than white, shall be produced by the use of vat dyes to match an approved standard shade or when required by the contracting officer, shall be produced by the use of such colorants or such methods as may be prescribed.

3.4.1 *Matching.*—The color shall match the approved standard shade under natural (north sky) daylight or artificial daylight having a color temperature of 7,500° Kelvin and shall be a good approximation to the standard shade under incandescent lamplight at 2,800° Kelvin.

3.4.2 *Colorfastness (except Air Force).*—The dyed and finished cloth shall show colorfastness to laundering, wet dry cleaning, perspiration, crocking and light equal to or better than the standard sample when tested as specified in 4.4. Unless otherwise specified (see 6.1 and 6.2), when no standard sample is available, the dyed and finished cloth shall show good colorfastness to laundering, wet dry cleaning, perspiration, crocking and light (40 standard fading hours), when tested as specified in 4.4.

3.4.2.1 *Colorfastness (Air Force only).*—For Air Force procurements, the colorfast requirements shall be as specified in the invitation for bids (see 6.2).

3.5 Physical requirements.—The finished cloth shall conform to the minimum requirements specified in table I when tested as specified (see 4.4).

TABLE I.—*Physical requirements*

Type	Class	Weight per sq. yard	Yarns per inch		Breaking strength (grab method)		Weave (for all types)
			Warp	Filling	Warp	Filling	
		Ounces			*Pounds*	*Pounds*	
I	1	7.7	72	60	130	90	3 harness, 2 left twill $\frac{2}{1}$
	2	7.5	74	58	120	85	
	3	6.5	74	56	105	70	
II	1	7.7	72	48	130	85	
	2	7.5	74	46	120	72	
	3	6.5	74	44	105	62	
III	1	6.7	72	48	110	65	
	2	6.5	74	46	100	60	
	3	6.0	74	44	90	55	

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:29Z
Check the source to verify that this is the current version before use.

3.5.1 *Width.*—The width shall be as specified (see 6.1).

3.6 Sizing, finishing and other nonfibrous materials.

3.6.1 *Class 1.*—The total sizing, finishing, and other nonfibrous materials shall not exceed 12 percent when tested as specified (see 4.4).

3.6.2 *Class 2.*—The cloth shall be desized and refinished so that the total sizing, finishing and other nonfibrous materials shall not exceed 5 percent when tested as specified (see 4.4).

3.6.3 *Class 3.*—The total finishing and other nonfibrous materials shall not exceed 2 percent when tested as specified (see 4.4).

3.7 Shrinkage.—Class 2 and class 3 cloths shall be preshrunk and shall not shrink more than 2 percent in the direction of the warp or more than 2 percent in the direction of the filling when tested as specified (see 4.4). The preshrinking process used shall not be identified by name or trade-mark, either on the cloth, ticket, or package.

3.8 Seam efficiency.—The cloth shall have a seam efficiency of not less than 80 percent when tested as specified (see 4.4).

3.9 Length.—The cloth shall be furnished in bolts of pieces having an average length of not less than 70 continuous yards with a minimum for any one piece of 40 continuous yards.

3.10 Tickets.—Each bolt shall have a ticket attached to the selvage at the inner end of the cloth with not finer than 5-ply cotton string doubled to not less than 8 inches long. The ticket shall be made of carboard not less than 0.020 inch in thickness and the color shall be white or light in intensity to permit easy reading of printed or penciled markings. The finish on the cardboard shall be suitable for pencil markings. The ticket shall have a reinforced eyelet for attaching the tying cord and shall be legibly printed with water insoluble ink with the inscription specified in Federal Specification PPP–P–51, except when other inscription data is specified by Civilian Government agencies.

3.11 Workmanship.—The cloth shall conform to the quality and grade of product established by this specification. The occurrence of defects shall not exceed the levels set by the applicable acceptable quality levels (see 4.2).

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 General.—The methods of sampling, inspection and testing contained in this section are applicable to inspection by and for the Government. Sampling, inspection and testing for components or materials described shall be in accordance with this specification and with subsidiary specifications referred to herein to the extent applicable, except that the quality assurance provisions of this specification shall govern in the event of conflict.

4.2 Preproduction inspection.—When specified, the preproduction sample submitted in accordance with 3.2 shall be visually inspected for appearance, color and finish. The sample shall be tested for physical and chemical properties in accordance with the methods shown in table IV, when applicable.

4.3 Inspection for acceptance. — Except when otherwise specified by Civilian Government agencies, inspection shall be in accordance with the provisions set forth in Military Standard MIL–STD–105 and Appendix except where otherwise indicated. An action by the Government shall, at the option of the Government, be performed either by the Government or the contractor under the supervision of the Government.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:29Z
Check the source to verify that this is the current version before use.

4.3.1 *Examination of the end item.*—Defects found during this examination shall be classified in accordance with 4.3.1.1, 4.3.1.2 and 4.3.1.3. The Government reserves the right to require examination for any defect not included herein and to classify such defects in accordance with the definition contained in Military Standard MIL–STD–105.

4.3.1.1 *Yard-by-yard examination.* — The defects listed in table II shall be counted regardless of their proximity to each other, except where two or more defects represent a single local condition of the cloth, in which case only the more serious defect shall be counted. A continuous defect shall be counted as one defect for each warpwise yard or fraction thereof in which it occurs. The unit of product for this examination shall be one linear yard. The inspection level shall be level II of Military Standard MIL–STD–105, and the acceptable quality levels shall be 1.5 majors and 6.5 total defects per 100 units (yards). The number of sample pieces over which the sample yardage shall be distributed shall be in accordance with table III.

TABLE II.—*Classification of defects*

Glossary No. Fed. Std. No. 4	Defect	Classification	
		Major	Minor
55, 56, 90, 91, 92	Hole, cut, tear or break (including selvage)—in which more than 3 yards are ruptured in adjoining points	X	
78	Smash—any	X	
78, 79	Abrasion mark, bruise, tender or weak spot—any	X	
11, 33	Broken or missing yarns—two or more contiguous, 2 inches or more in length	X	
	Two contiguous less than 2 inches in length, or single yarn missing for more than 9 inches		X
9	Fine or light filling bar, light place clearly noticeable	X	
1, 2, 4, 26	Coarse filling bar, heavy place, double or coarse yarns—clearly noticeable		X
3, 5, 7, 27, 29, 77	Slough-off, jerked-in filling, slubs, gouts, surface knots—more than three times the thickness of the normal yarns		X
12, 35	Open place, crack or hitchback—clearly noticeable	X	
13, 34	Floats, multiple ½ inch or more in combined warp and filling direction	X	
	Multiple less than ½ inch in combined warp and filling directions, or single 9 inches or more in length		X
	Skips (harness)—clearly noticeable	X	
50	Tight selvage—resulting in pronounced slackness or waviness in the cloth		X
14	Shade bar or mixed filling—clearly noticeable at distance of 6 feet		X
69, 70, 74, 75, 76	Spot, stain or streak—any uncleaned		X
59, 61	Crease or wrinkle—embedded, clearly noticeable		X

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:29Z
Check the source to verify that this is the current version before use.

TABLE III.—*Sampling table*

Lot size in yards	Sample size in pieces	Acceptance number
Up to 1,300[1]	3	0
1,301 to 3,200	5	0
3,201 to 8,000	7	0
8,001 to 22,000	10	0
22,001 to 110,000	15	1
110,001 to 550,000	25	1
550,001 and over	50	2

[1] If lot contains fewer than 3 pieces, each piece in the lot shall be examined.

TABLE IV.—*Test methods*

Characteristic	Requirement reference	Test method
Total non-fibrous material	3.6	2611
Weight, oz. per square yard	Table I	5041
Yarn per inch	Table I	5050
Breaking strength	Table I	5100
Seam efficiency[1]	3.8	5110
Shrinkage in laundering	3.7	5550
Colorfastness:		
Laundering	3.4.2	5610
Wet dry cleaning	3.4.2	5622
Crocking	3.4.2	5650
Light	3.4.2	5660
Perspiration[2]	3.4.2	5680

[1] The needle and thread size conforming to Federal Specification V–T–276 shall be as follows:

Needle size 0.040 (± 0.001) inch diameter across the blade, round point.

Thread—Top 40/3
Looper 60/3

[2] Except that the transfer cloth shall be a 6-fiber cloth containing 2-inch warp of wool, spun nylon, spun rayon, spun silk, spun acetate and bleached cotton crossed with filling of orlon 81.

4.3.1.2 *Over-all examination.*—Each defect listed below shall be counted no more than once in each piece examined. The unit of product for this examination shall be one piece. The sample size (number of pieces selected as sample) for this examination and the acceptance number (maximum number of defects acceptable) shall be as shown in table III.

Defects

Width less than specified.

Baggy, ridgy or wavy cloth, clearly noticeable.

Selvage curled, doubled, folded or rolled intermittently, or continuously over more than 20 yards of the piece.

Off-shade—not within established tolerance.

Nonuniform in shade—shaded end to end, side to center, side to side, overall uncleanliness.

Objectionable odor.

4.3.1.3 *Length examination.*

4.3.1.3.1 *Individual pieces.*—The piece shall be considered defective with regard to length requirements if its actual gross length is less than 40 yards, or more than 2 yards shorter than the gross length marked on the piece ticket. The sample size (number of pieces selected on sample) and acceptance number (maximum number of defects acceptable) for this examination shall be as shown in table III.

4.3.1.3.2 *Total yardage in sample.*—The lot shall be unacceptable if the total of the actual gross length of pieces in the sample is less than the total of the gross lengths marked on piece tickets, or the average piece length is less than 70 yards.

4.3.2 *Inspection for chemical and physical tests.*—For each characteristic the acceptable quality level shall be 4.0 per hundred units, and the inspection level shall be level L–1, table VII of Appendix to Military Standard MIL–STD–105. The lot size shall be expressed in units of 1 yard each.

4.4 **Tests.**—The methods of testing described in Federal Specification CCC–T–191 wherever applicable and as listed in table IV, shall be followed. The physical or chemical

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:29Z
Check the source to verify that this is the current version before use.

values specified in section 3 represent the average of the number of test specimens selected for a unit of product for test purposes as specified in the applicable test methods. No unit of product value shall fall below the minimum or above the maximum specified. The unit of product for testing shall be three full widths continuous yards of the cloth.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking.—Except when otherwise specified by Civilian Government agencies, the cloth shall be packaged, packed and marked in accordance with class 3 of Federal Specification PPP–P–51. Packaging and packing shall be level A, B, or C as specified in the contract or order. (See 6.1 and 6.1.1.)

5.1.1 *Federal Standard No. 102.*—Federal Standard No. 102 shall be referred to for definitions and applications of the various levels of packaging protection for supplies and equipment.

6. NOTES

6.1 Ordering data. — Procurement documents should specify the following:

(*a*) Title, number and date of this specification.

(*b*) Type and class of cloth required. (see 1.2).

(*c*) Availability and requirements for standard sample (see 3.1).

(*d*) Requirements for preproduction sample approval (see 3.2).

(*e*) Color and colorfastness requirements (when required) (see 3.4, 3.4.2 and 3.4.2.1).

(*f*) Width of cloth required (see 3.5.1).

(*g*) Inscription data required on ticket (see 3.10).

(*h*) Both the level of packaging and the level of packing required. (see 5.1 and 6.1.1).

6.1.1 In the preparation of contracts or orders, it must be noted that various options, choices and alternatives, as indicated in Federal Specification PPP–P–51, may be exercised in the preparation for delivery of this commodity.

6.2 Air Force. — For Air Force procurements, the color and colorfastness requirements shall be specified by the procuring activity.

6.3 This specification includes the requirements of Military Specification MIL–C–300B.

6.4 Transportation description.—The transportation description applicable to this item is:

Cotton piece goods, not otherwise indexed by name in the original piece.
Carload minimum weight—any quantity.
Motor volume, minimum weight—any quantity.

Notice. — When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility, nor any obligation whatsoever and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise, as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

MILITARY INTERESTS:
Army—Q M
Navy—S MC Sh
Air Force.

Copies of this specification may be purchased for 10 cents each.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:29Z
Check the source to verify that this is the current version before use.